Power equations

1 Solve the following equations:

a $x^5 = 3^5$

b $8^{-7} = x^{-7}$

c $x^3 = \left(\frac{8}{5}\right)^3$

d $x^{-7} = \frac{1}{6^7}$

e $3\left(x^{-9}\right) = \frac{3}{2^9}$

f $x^{\frac{1}{3}} = \sqrt[3]{6}$

g $\sqrt[3]{5} = x^{\frac{1}{3}}$

h $\frac{1}{2^5} = x^{-5}$

Exponential equations

2 Solve the following equations:

a $4^x = 4^8$

b $9^x = 9^{-4}$

c $3^x = 3^{\frac{2}{9}}$

d $3^x = 27$

e $7^x = 1$

f $10^x = 0.01$

g $\left(\frac{8}{3}\right)^x = \left(\frac{8}{3}\right)^7$

h $8^x = \frac{1}{8^2}$

i $7\left(4^x\right) = \frac{7}{4^3}$

j $5^x = \sqrt[3]{5}$

k $8^x = \sqrt{8}$

l $30^n = \sqrt[3]{30}$

3 Find the interval, of two consecutive integers, in which the solution of the following equations will lie:

a $3^x = 57$

b $3^x = 29$

c $2^x = \frac{1}{13}$

d $2^x = -5$

4 Consider the following equations:

i Rewrite each side of the equation with a base of 2.

ii Hence, solve for x.

a $8^x = 4$

b $16^x = \frac{1}{2}$

c $\frac{1}{1024} = 4^x$

d $\left(\sqrt{2}\right)^x = \sqrt[5]{32}$

5 Solve the following exponential equations:

a $9^y = 27$

b $9^{x+3} = 27^x$

c $\left(\sqrt{6}\right)^y = 36$

d $\left(\sqrt{2}\right)^k = 0.5$

e $3^{5x-10} = 1$

f $5^{-3x-1} = 3125$

g $5^{10x+33} = 125$

h $\dfrac{1}{3^{x-3}} = \sqrt[3]{9}$

i $\left(\dfrac{1}{9}\right)^{x+5} = 81$

j $\left(\dfrac{1}{8}\right)^{x-3} = 16^{4x-3}$

k $\dfrac{25^y}{5^{4-y}} = \sqrt{125}$

l $8^{x+5} = \dfrac{1}{32\sqrt{2}}$

m $30 \times 2^{x-6} = 15$

n $2^x \times 2^{x+3} = 32$

o $\left(2^2\right)^{x+7} = 2^3$

p $\left(2^4\right)^{2x-10} = 2^2$

q $81^{x-1} = 9^{3x+5}$

r $25^{x+1} = 125^{3x-4}$

s $a^{x-1} = a^4$

t $a^{x+1} = a^3\sqrt{a}$

u $3^{x^2-3x} = 81$

v $27\left(2^x\right) = 6^x$

6 For each of the following equations:

i Simplify the left side of the equation.

ii Solve the equation for x.

a $3^x \times 3^{nx} = 81$

b $3^x \times 9^{x-k} = 27$

7 Consider the following equations:

i Determine the substitution, m that would reduce the equation to a quadratic.

ii Hence, solve the equation for x.

a $\left(2^x\right)^2 - 9 \times 2^x + 8 = 0$

b $2^{2x} - 12 \times 2^x + 32 = 0$

c $4 \times 2^{2x} - 34 \times 2^x + 16 = 0$

d $4^{2x} - 65 \times 4^x + 64 = 0$

e $4^x - 5 \times 2^x + 4 = 0$

f $9^x - 12 \times 3^x + 27 = 0$

8 Given that $2^x \times 5^{x+2} = 50$, find the value of $2^{x+2} \times 5^x$.

Applications

9 Find the x-coordinate of the point of intersection of the graphs of $y = 2^{5x}$ and $y = 4^{x-3}$.

10 Find the value of h, given the point $\left(h, \frac{1}{9}\right)$ lies on the curve $y = 3^{-x}$.

11 Given the points $(3, n)$, $(k, 16)$ and $\left(m, \frac{1}{4}\right)$ all lie on the curve with equation $y = 2^x$, find the value of:

a n

b k

c m

12 A certain type of cell splits in two every hour and each cell produced also splits in two each hour. The total number of cells after t hours is given by:

$$N(t) = 2^t$$

Find the time when the number of cells will reach the following amounts:

a 32

b 1024

c 4096

13 The frequency $f\ (\mathrm{Hz})$ of the nth key of an 88-key piano is given by $f(n) = 440\left(2^{\frac{1}{12}}\right)^{n-49}$.

a Find the frequency of the forty-ninth key.

b Find the frequency of the 40th key to the nearest whole number.

c Find the value of n that corresponds to the key with a frequency of $1760\ \mathrm{Hz}$.